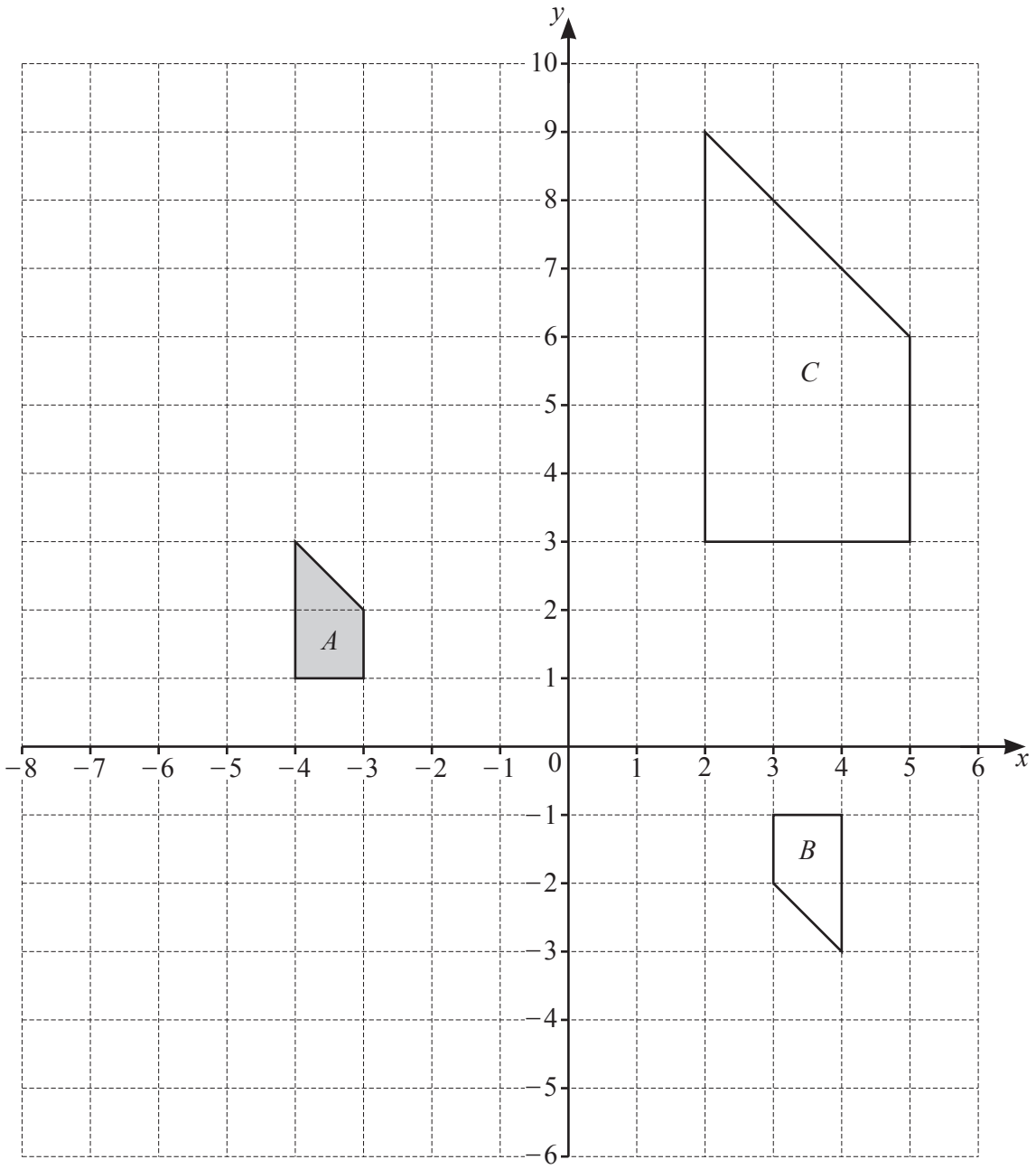
0580_w22_qp_33_Q8

8 (a)



(i) Draw the image of shape A after a reflection in the line $y = -1$. [2]

(ii) Describe fully the **single** transformation that maps shape A onto shape B.

.....

..... [3]

(iii) Describe fully the **single** transformation that maps shape A onto shape C.

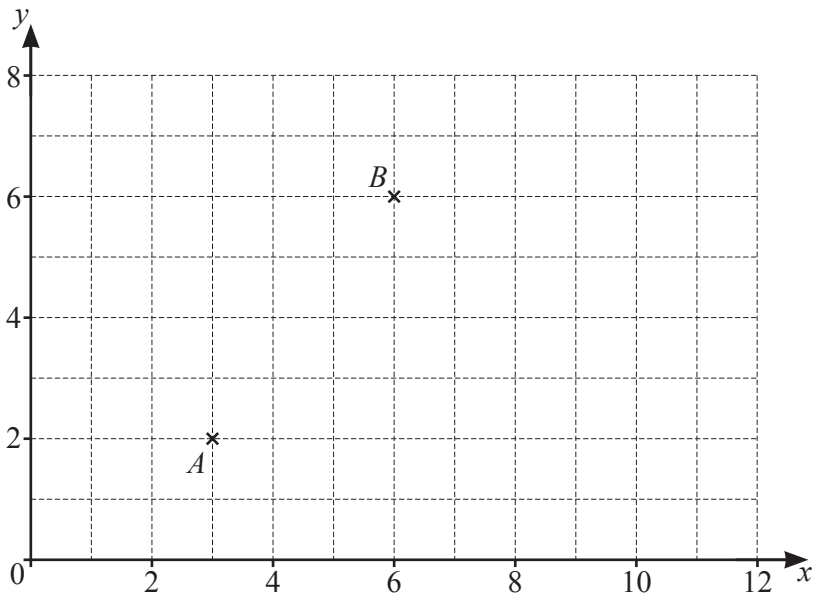
.....

..... [3]

(iv) Complete this statement.

The area of shape C is ..... times bigger than the area of shape A. [2]

(b)



(i) Write $\overrightarrow{AB}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ [1]

(ii) $\overrightarrow{BC} = \begin{pmatrix} 4 \\ -5 \end{pmatrix}$

On the grid, plot point C. [1]

(c) $\mathbf{p} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$ $\mathbf{t} = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$

Work out

(i) $3\mathbf{p}$,

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ [1]

(ii) $\mathbf{t} - \mathbf{p}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ [1]